DEEP LEARNING PROJECT

CIFAR-10 Image Classification

From a shallow CNN baseline to residual networks and transfer learning

Team

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Università degli Studi di Milano-Bicocca · A.A. 2024/2025 · Framework: TensorFlow / Keras

OVERVIEW

What we'll cover

The problem

Why CIFAR-10 is a meaningful benchmark

The data

60k images, 10 classes, augmentation strategies

Architectures

Shallow, Deep, Residual and Transfer Learning families

Hyperparameters

Does targeted hyperparameter tuning improve model performance?

Results

Quantitative and qualitative analysis across all models

Conclusions

Answers to our five research questions

THE PROBLEM

Classifying small, noisy images into 10 categories

CIFAR-10 is a long-standing computer vision benchmark: 60,000 32×32 colour images split across 10 everyday object classes. Despite their tiny size, the images contain enough visual signal to separate similar categories — but also enough ambiguity (cat vs. dog, deer vs. horse) to make the task a genuine test of representation learning.

Goal

Build and compare multiple CNN architectures of increasing complexity — and explore targeted hyperparameter choices — to evaluate what most improves generalization on this benchmark.

Research Questions

- RQ1 — Can a custom CNN classify CIFAR-10 effectively?
- RQ2 — Does targeted hyperparameter tuning improve model performance?
- RQ3 — Does data augmentation reduce overfitting?
- RQ4 — Which classes are most often confused?
- RQ5 — Does transfer learning beat custom CNNs?

THE DATA

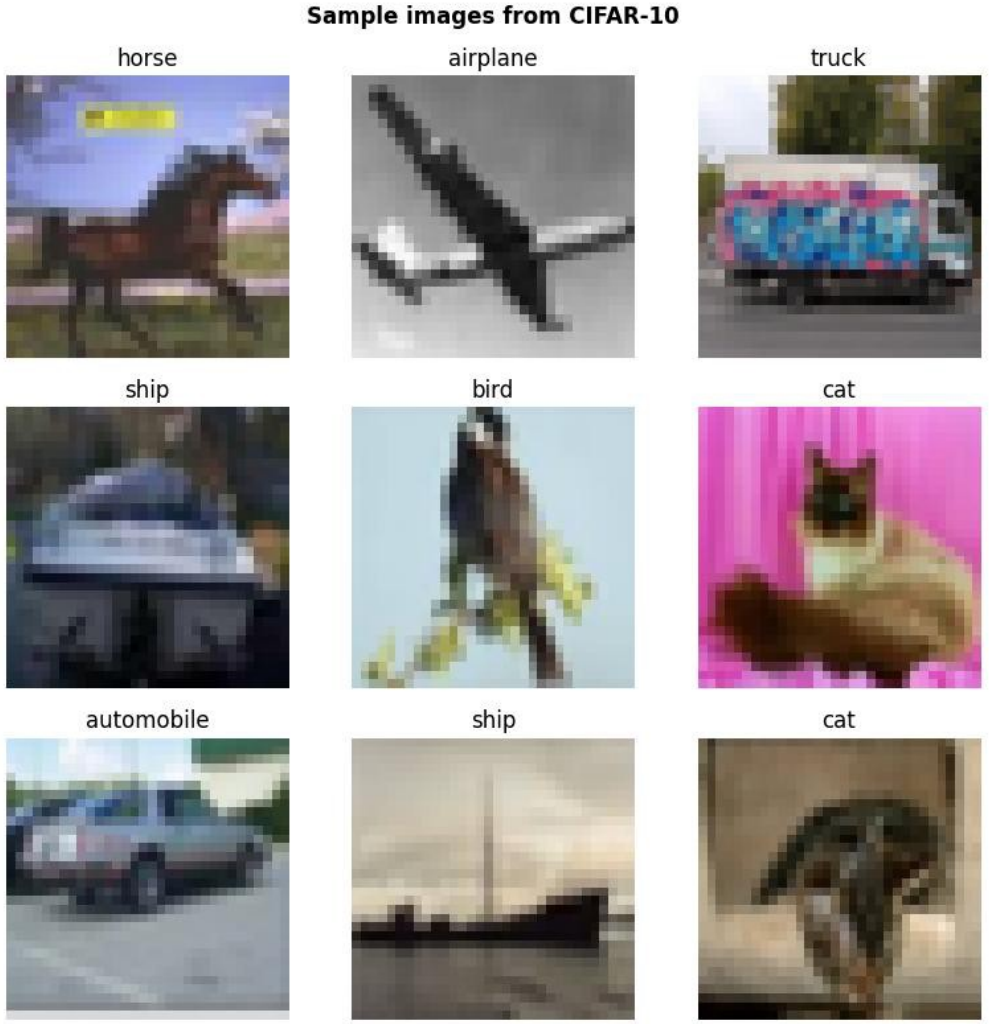
CIFAR-10 at a glance

60,000 — total RGB images

10 — balanced classes

32×32 — pixels per image

45k / 5k / 10k — train / val / test split



THE DATA

Preprocessing and data augmentation

Pixel values are rescaled and per-channel normalized inside the model graph (Keras Rescaling + Normalization layers), so the same preprocessing applies at both train and inference time.

Two augmentation regimes were compared across multiple architectures:

Standard augmentation

- Random horizontal flip
- Small translation
- Small contrast variation

Stronger augmentation

- Random horizontal flip
- Wider translation
- Random zoom
- Stronger contrast variation

⚠️ Stronger augmentation often hurt accuracy — too much distortion at 32×32 resolution.



Same image, 9 augmented variants

OUR SOLUTION

Four architecture families, multiple variants



15 models trained in total. All share the same 45k/5k/10k split and the same evaluation metrics — differences reflect design and hyperparameter choices, not pipeline differences.

🏆 **Best result: Residual CNN — 91.6% test accuracy**

SOLUTION — MODELS 1 & 1B

Shallow CNN: baseline and augmented variant

Two conv + max-pool blocks (32 → 64 filters), Dense(256) head with Dropout. ~1.07M parameters. Same architecture trained twice: without augmentation (Model 1) and with standard augmentation (Model 1b).

72.8%

Model 1 — test acc

79.7%

Model 1b + Aug — test acc

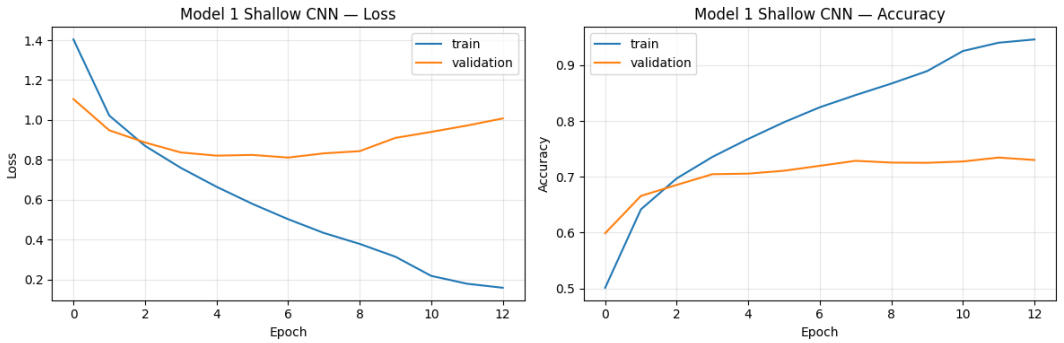
0.820

Model 1 — test loss

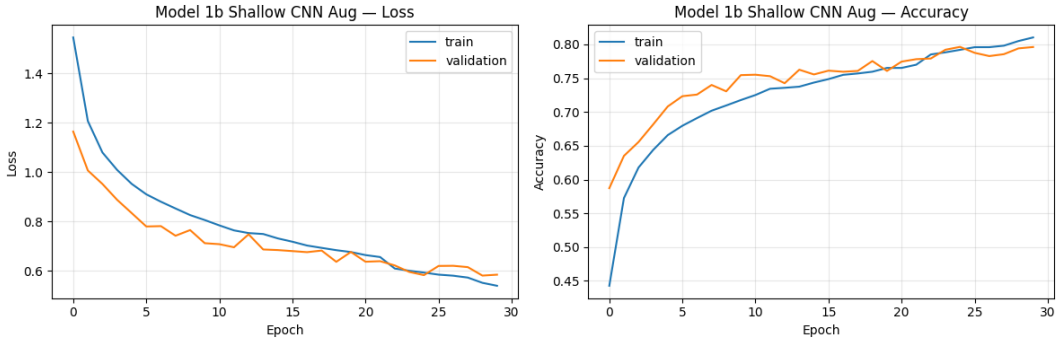
0.602

Model 1b — test loss

Augmentation alone: +6.9 pts accuracy, train/val gap closed → direct answer to RQ3



Model 1 — Shallow CNN (baseline)



Model 1b — Shallow CNN + Augmentation

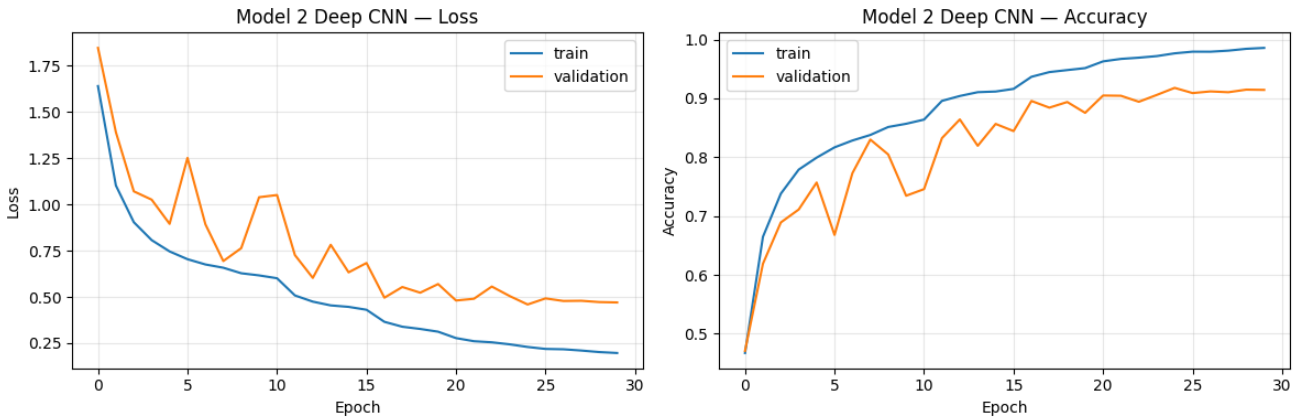
SOLUTION — MODEL 2

Deep CNN: architecture and targeted hyperparameter tuning

Eight conv layers in four blocks (64→128→256→256 filters) with BatchNorm, GlobalAveragePooling and a dropout head (~2.47M params). Five training variants explored:

Variant	Test Acc	Test Loss	Observation
Deep CNN (lr=1e-3)	**91.2%**	0.466	Scheduler-driven LR decay — best Deep CNN variant
Deep CNN (lr=5e-4)	**89.5%**	0.472	Fixed lower LR, no scheduler — slightly worse
+ clipnorm	**89.9%**	0.488	Gradient clipping adds stability, minor accuracy cost
+ L2 2e-4	**89.2%**	0.535	Extra L2 regularization over-constrains the model
+ Dense 256	**90.2%**	0.466	Smaller 256-unit head; competitive result

Deep CNN: training dynamics and per-class results



Training & validation — Deep CNN lr=1e-3

91.2%
test accuracy

0.466
test loss

2.47M
parameters

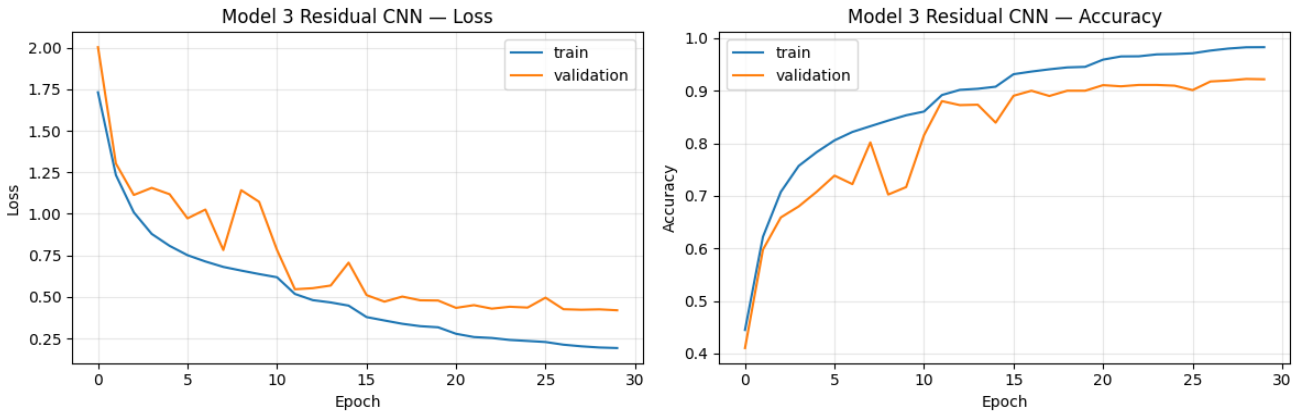
+18.4 pts over shallow baseline — direct answer to RQ2

Per-class F1 score — Deep CNN

airplane	0.920	automobile	0.963
bird	0.885	cat	0.825
deer	0.907	dog	0.863
frog	0.920	horse	0.937
ship	0.950	truck	0.949

Weakest: cat (0.825) and dog (0.863)

Residual CNN: training dynamics and per-class results



Training & validation — Residual CNN lr=1e-3

91.6%
test accuracy — best

0.420
test loss

2.78M
parameters

+0.4 pts over Deep CNN — skip connections ease gradient flow

Per-class F1 score — Residual CNN

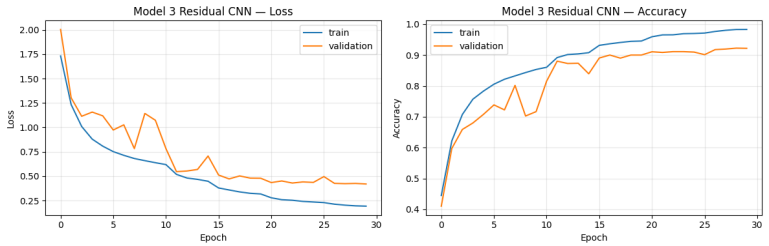
airplane	0.927	automobile	0.960
bird	0.877	cat	0.837
deer	0.916	dog	0.866
frog	0.931	horse	0.952
ship	0.951	truck	0.943

Best model overall — all classes above 0.837

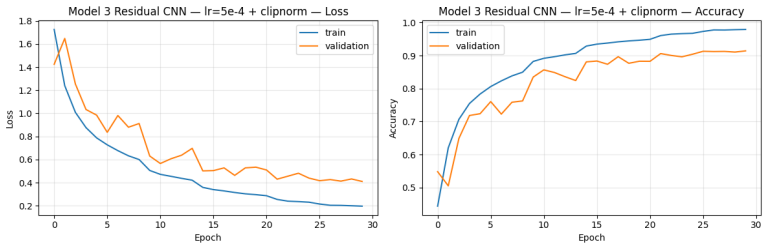
SOLUTION — MODEL 3

Residual CNN: ablation variants

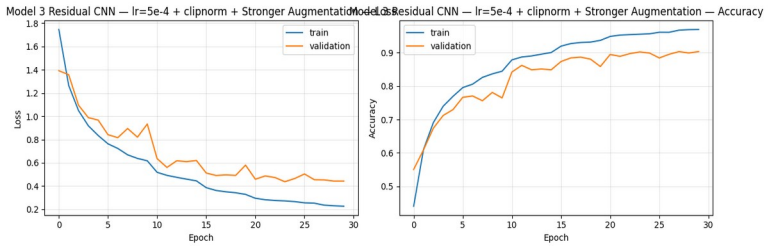
Variant	Test Acc	Test Loss	Key finding
Residual CNN (lr=1e-3)	**91.6%**	0.420	Best overall — default scheduler + standard aug
+ lr=5e-4 + clipnorm	**91.2%**	0.419	Stable training, −0.4 pts; clips gradient norm to 1.0
+ Stronger Augmentation	**90.2%**	0.453	Wider transforms destroy fine detail at 32×32



lr=1e-3 (91.6%, loss 0.420)



lr=5e-4 + clipnorm (91.2%, loss 0.419)



Stronger Aug (90.2%, loss 0.453)

Stronger augmentation likely removes useful fine details at 32×32.

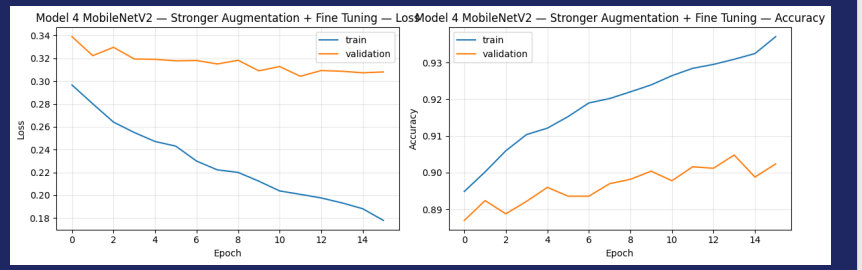
SOLUTION — MODELS 4 & 5

Transfer learning: MobileNetV2 and ResNet50

ImageNet-pretrained backbones (frozen) with custom heads. Five variants tested — fine-tuning, stronger augmentation and backbone size:

Model	Acc	Loss	Observation
MobileNetV2 (frozen)	88.4%	0.341	Competitive frozen baseline
MobileNetV2 Fine-tuned	89.0%	0.365	Partial unfreeze: +0.6 pts
MobileNetV2 + Stronger Aug	88.3%	0.349	Stronger aug hurts here too
MobileNetV2 Aug + Fine-tuned	**90.0%**	0.320	Best transfer learning result
ResNet50 (frozen)	87.7%	0.371	Larger backbone — head undertrained

MobileNetV2 Aug + Fine-tuning



Training & validation — fine-tuning phase

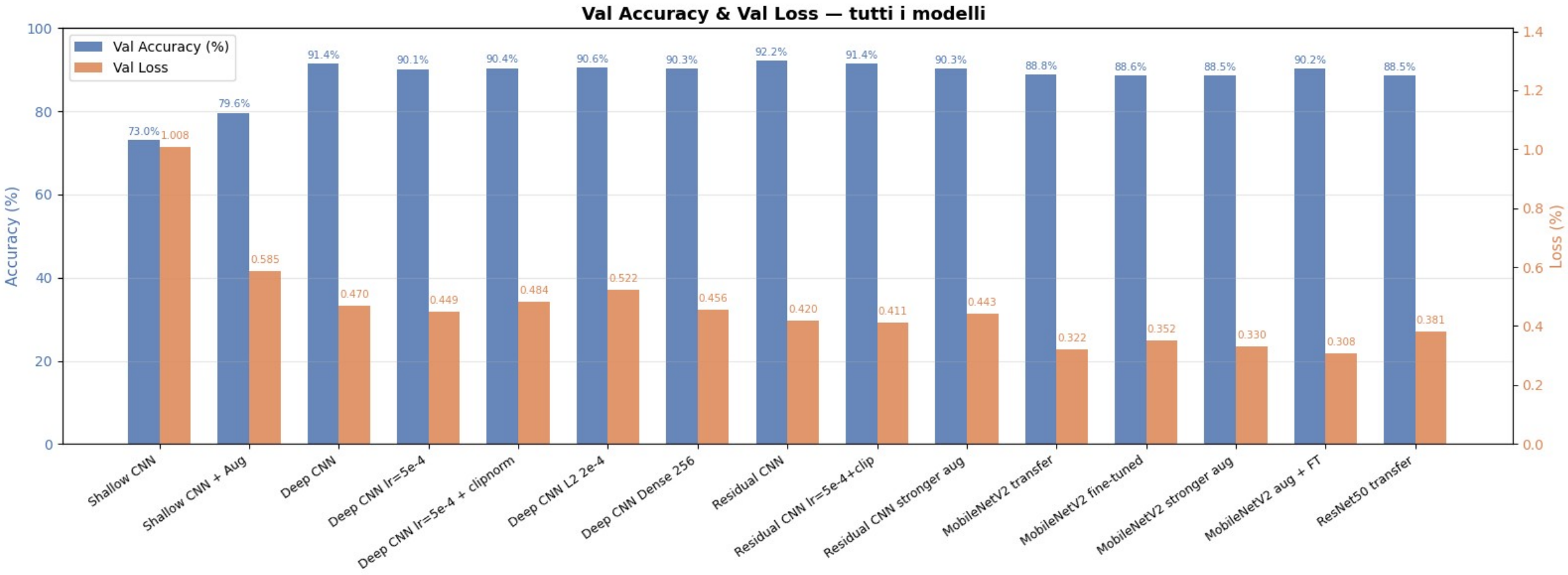
90.0%
test accuracy

0.320
test loss

Best transfer learning result

RESULTS

Validation accuracy & loss — all models



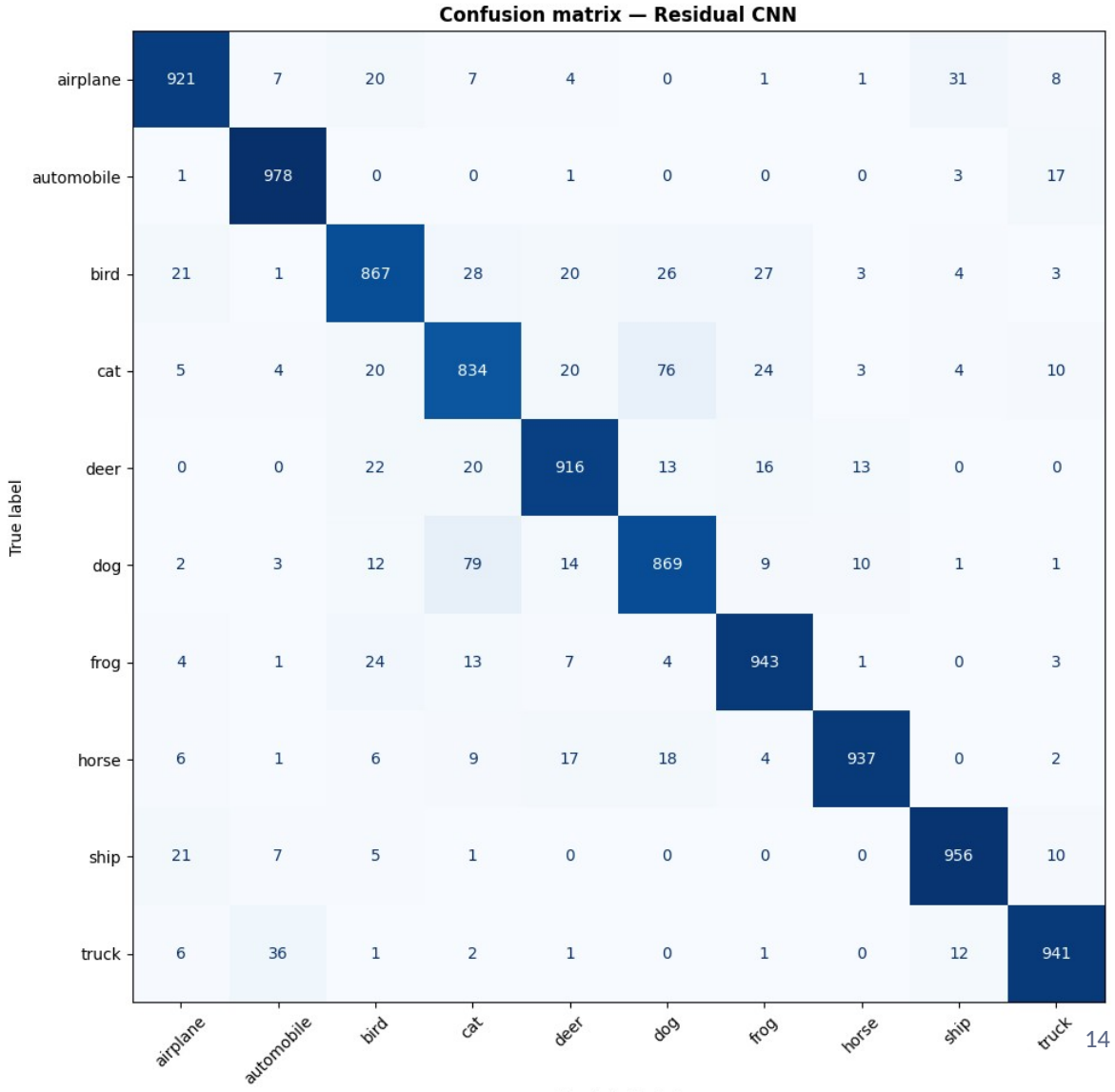
RESULTS — RQ4

Confusion matrix: Residual CNN (91.6%)

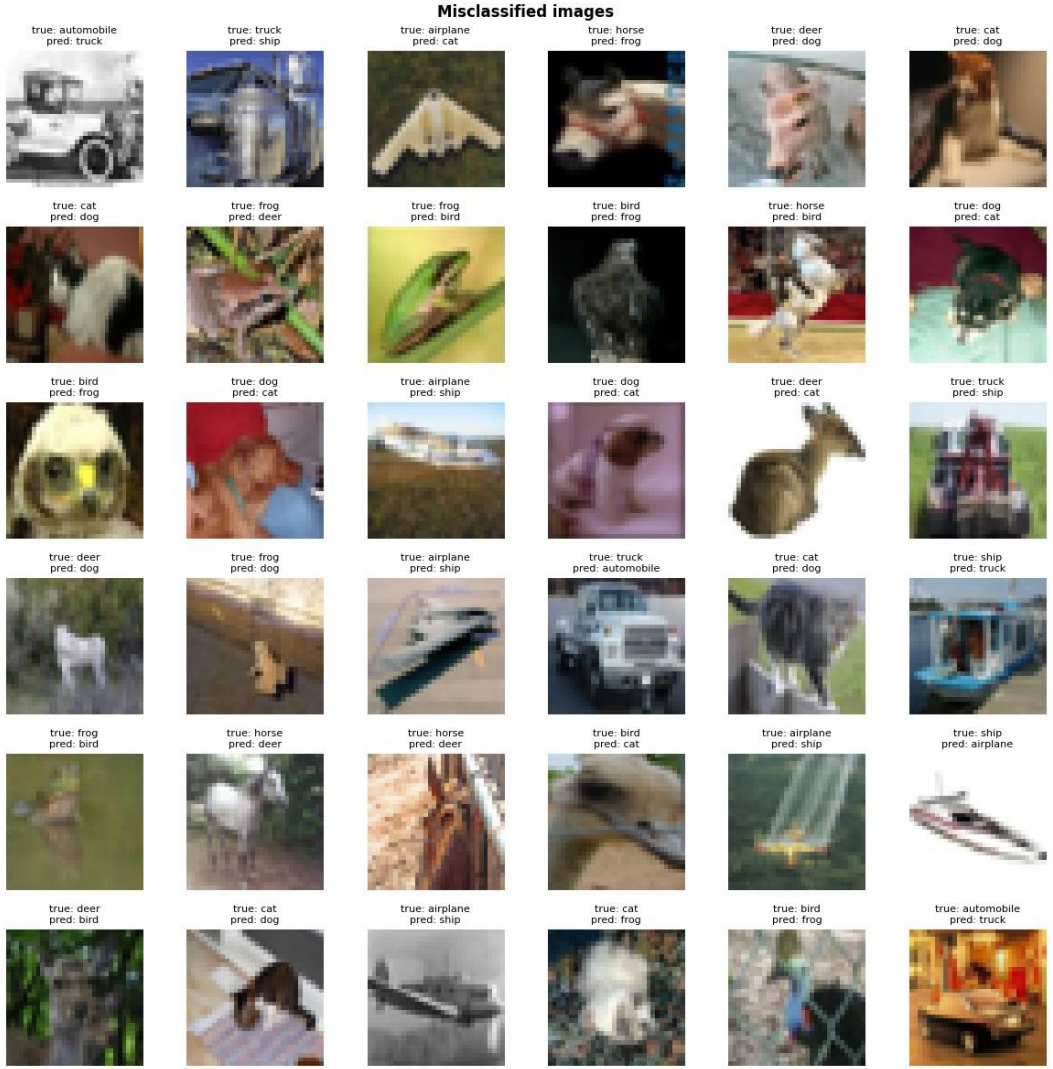
Using the best model, errors concentrate on visually similar pairs:

- cat ↔ dog — 76 + 79 confusions, largest pair
- bird ↔ frog / cat — small textured animals
- deer ↔ horse — similar four-legged silhouettes
- automobile ↔ truck — both wheeled vehicles
- ship ↔ airplane — large objects, similar colour

Residual CNN — test accuracy 91.6% · test loss 0.420



Misclassified images and what they reveal



What the errors look like

Most misclassified images are genuinely ambiguous at 32×32: low light, unusual poses, occlusion or very low contrast. Several errors are defensible even to a human viewer.

- cat ↔ dog and deer ↔ horse are the most frequent confusion pairs — visually similar at this resolution.
- Textured animals (bird, frog) are often confused with each other and with cats.
- Vehicles (automobile ↔ truck, airplane ↔ ship) share shape and colour cues.
- Low-light and partial-occlusion images are hard even for humans — these errors are expected.
- The best model (Residual CNN, 91.6%) still misclassifies ~840 out of 10,000 test images.

CONCLUSIONS

Answering our research questions

RQ1 Yes — a simple CNN reaches 72.8%. Standard augmentation alone lifts it to 79.7%.

RQ2 Partially — targeted hyperparameter changes affected stability and loss, but did not consistently improve accuracy. The default lr=1e-3 setup with ReduceLROnPlateau remained the strongest configuration.

RQ3 Yes — standard augmentation reduces overfitting. Stronger augmentation alone often hurts at 32×32, although it helped MobileNetV2 when combined with fine-tuning.

RQ4 Cat/dog, deer/horse and automobile/truck are the dominant confusions — all visually similar pairs.

RQ5 Partially — MobileNetV2 Aug + Fine-tuned reaches 90.0%, ResNet50 87.7%; both trail the best custom CNN (91.6%).

🏆 *Best model: Residual CNN — 91.6% test accuracy · Università degli Studi di Milano-Bicocca, A.A. 2024/2025*